

1.1.5

```
1. module pr-encoder(  
    input [3:0] D,  
    output reg [1:0] out,  
    output reg v  
);
```

```
    always @(*) begin
```

```
        if (D == 4'b0000) begin
```

```
            out = 2'b00;
```

```
            v = 1'b0;
```

```
        end
```

```
        else if (D[3] == 1) begin
```

```
            out = 2'b11;
```

```
            v = 1'b1;
```

```
        end
```

```
        else if (D[2] == 1) begin
```

```
            out = 2'b10;
```

```
            v = 1'b1;
```

```
        end
```

```
else if (D[1]==1) begin
```

```
out = 2'b01;
```

```
v = 1'b1;
```

```
end
```

```
else if (D[0]==1) begin
```

```
out = 2'b00;
```

```
v = 1'b1;
```

```
end
```

```
else begin
```

```
out = 2'b00;
```

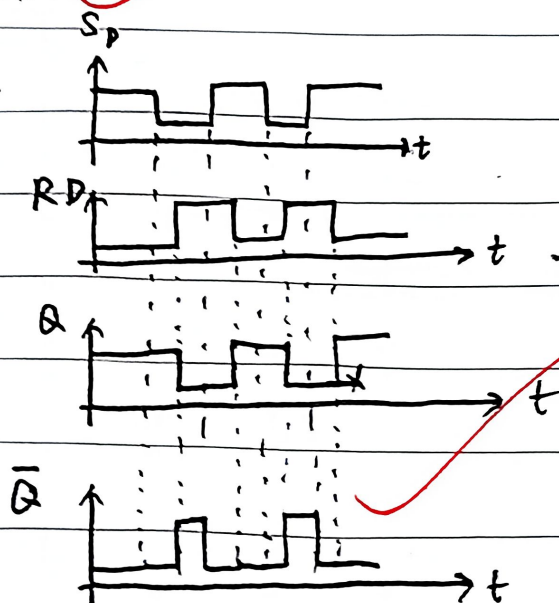
```
v = 1'b0;
```

```
end
```

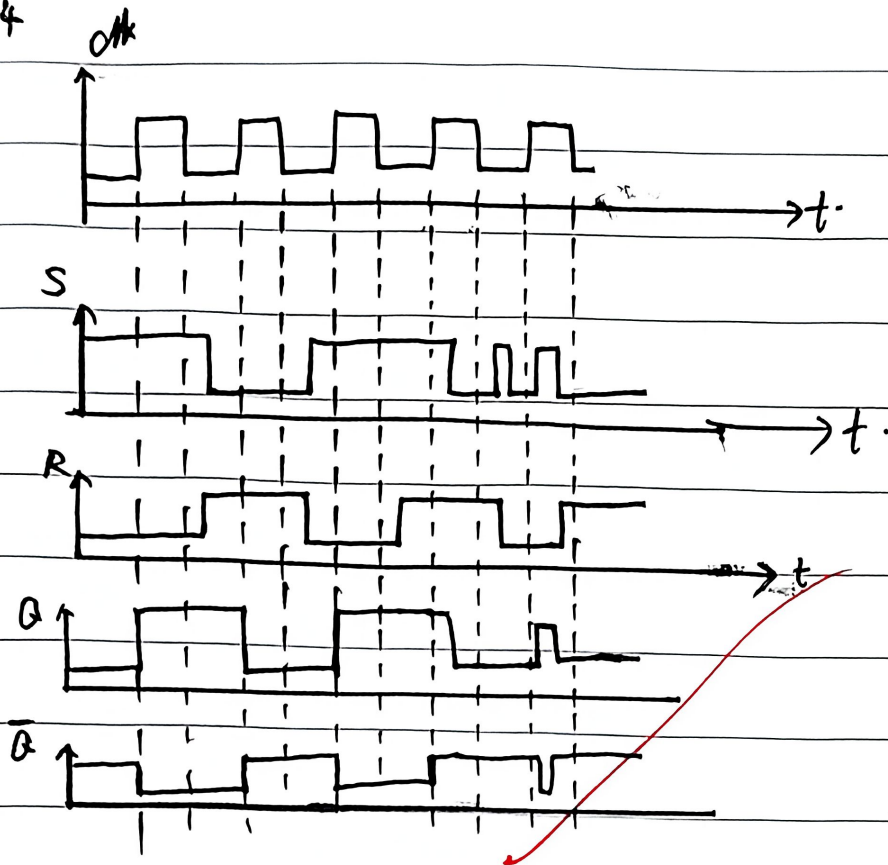
```
end
```

```
endmodule
```

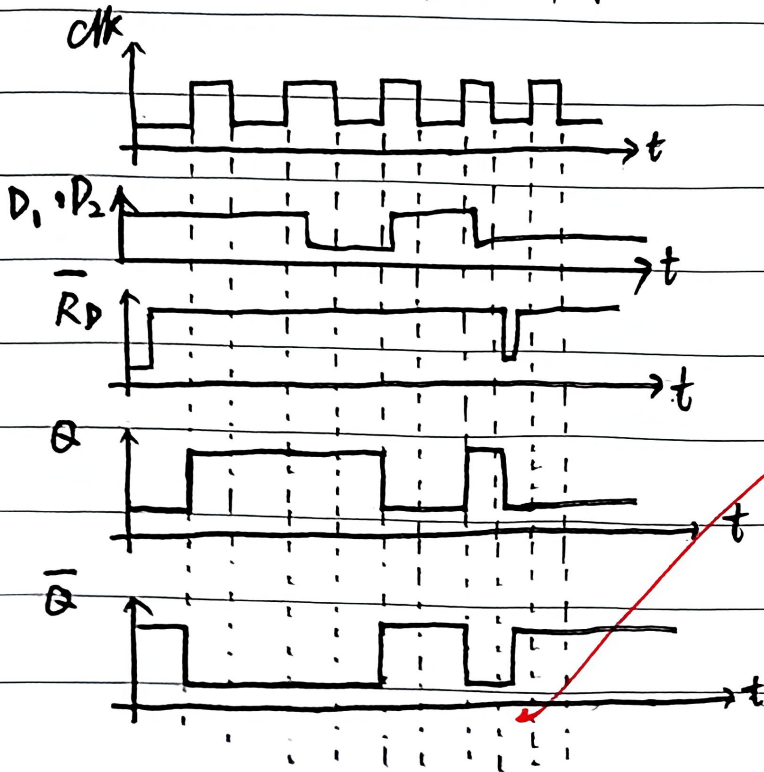
2. 5.2



5.4



5.8 上升沿触发 R_D 异步低电平有效



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